Question	Answer	Marks	Partial Marks	Guidance
4(a)	250	**2**	**B1** for 6750 or **M1** for 7000 – 50 × 135 or for 7 – 50 × 0.135	 Implied by answer 0.25
4(b)	80	**2**	**M1** for 6 × 6 × 4 or for 4^3 oe OR **M1** for (6 × 6) – (4 × 4) oe	Implied by 144 or by 64 Implied by 20 oe eg 6 × (6 – 4) + (6 – 4) × 4
4(c)	118 or 117.7 to 117.8	**4**	**M3** for $\frac{1}{2}\times 4\times 4\times \sin 60$ × 20 × 0.85 oe OR **M1** for $\frac{1}{2}\times 4\times 4\times \sin 60$ or $\frac{1}{2}\times 4\times \sqrt{4^2-2^2}$ oe **M1** for 20 × *their* area of triangle **M1 dep** for 0.85 × *their* volume, dependent on previous M1 If 0 scored **SC1** for height = 3.46...	M3 implied by $68\sqrt{3}$. 6.928... or $4\sqrt{3}$ *Their* area must be from trig or Pythag. or from $\frac{1}{2}\times 4\times 4$. Implied by 138.56... or $80\sqrt{3}$ Accept $2\sqrt{3}$

Question	Answer	Marks	Partial Marks	Guidance
4(d)(i)	$\pi\times\sqrt{24^2+10^2}\times10+\pi\times10^2$ or $\pi\times\left(\sqrt{24^2+10^2}\right)^2\times\frac{2\times\pi\times10}{2\times\pi\times26}+\pi\times10^2$	**M3**	**M2** for $\pi\times\sqrt{24^2+10^2}\times10$ or $\pi\times\left(\sqrt{24^2+10^2}\right)^2\times\frac{2\times\pi\times10}{2\times\pi\times26}$ or **M1** for 24^2+10^2 or $\pi\times10^2$	M3 implied by 360π. M2 implied by 260π or 816.8 to 816.92. M1 implied by 576 or 100π oe. Allow use of $\frac{22}{7}$ or 3.14 for all M marks.
	1130.9 to 1131.1...	**A1**	Must see at least 5 sf	
4(d)(ii)(a)	0.151 or 0.1511 to 0.1512...	**1**		
4(d)(ii)(b)	22	**2**	**B1** for figs 22[1…] or **M1** for $\frac{2.5\times100^2}{1131}$	Allow more accurate value of 1131